

WireCell Toolkit Build Cheat Sheet

(ICARUS / LArSoft Environment) — updated for WCT master

Environment Setup

```
cd /home/usher/test/wirecell/wire-cell-toolkit
git checkout -- wscript          # ensure clean build script
# Core environment
setup gcc      v12_1_0
setup python v3_9_15
setup root     v6_28_12 -q e26:prof:p3915
# Dependencies
setup spdlog v1_14_1 -q e26:prof # <-- CHANGED: external-fmtlib build
setup fmt     v11_0_2 -q e26:prof # <-- ADDED:  required by spdlog v1_14_1
setup libtorch v2_1_1b -q e26     # <-- ADDED:  builds the WireCellPytorch plugin
setup boost    v1_82_0  -q e26:prof
setup gojsonnet v0_18_0 -q e26
setup jsoncpp  v1_9_5a  -q e26:prof
setup fftw     v3_3_10
setup hdf5     v1_12_2a -q e26:prof
setup protobuf v3_21_12a -q e26
```

Fix for issue #491 build error “#error WCT requires SPDLOG to be compiled against external fmtlib”. WCT master now requires SPDLOG built against *external* fmtlib. spdlog v1.9.2 bundles fmt and must be replaced by spdlog v1.14.1 + fmt v11.0.2.

ROOT Setup

```
export ROOTSYS=$(root-config --prefix)
export PATH=$ROOTSYS/bin:$PATH
export LD_LIBRARY_PATH=$(root-config --libdir):$LD_LIBRARY_PATH
export LIBRARY_PATH=$(root-config --libdir):$LIBRARY_PATH
export CPATH=$(root-config --incdir):$CPATH
```

Jsonnet Link Workaround

```
export GOJSONNET_DIR=/cvmfs/larsoft.opensciencegrid.org/products/gojsonnet/v0_18_0/Linux64bit+3.10-2.17-...
export LIBRARY_PATH=$GOJSONNET_DIR/lib:$LIBRARY_PATH
export LD_LIBRARY_PATH=$GOJSONNET_DIR/lib:$LD_LIBRARY_PATH
export CPATH=$GOJSONNET_DIR/include:$CPATH
export LDFLAGS="-L$GOJSONNET_DIR/lib $LDFLAGS"
export CXXFLAGS="-I$GOJSONNET_DIR/include $CXXFLAGS"
```

Local pkg-config Overrides

```
export PKG_CONFIG_PATH=/home/usher/test/wirecell/pkgconfig:$PKG_CONFIG_PATH
```

Updated: the cvmfs .pc files carry a stale prefix=/scratch/workspace/... and spdlog’s must advertise external fmtlib. Put corrected spdlog.pc and fmt.pc (provided alongside this sheet) in the override dir. Verify:

```
pkg-config --cflags spdlog      # MUST contain -DSPDLOG_FMT_EXTERNAL and an -I.../fmt/.../include
```

Boost Configuration

```
BOOST_DIR=/cvmfs/larsoft.opensciencegrid.org/products/boost/v1_82_0/Linux64bit+3.10-2.17-e26-prof
```

Configure / Build / Install

```
./wcb configure \
  --prefix=/home/usher/test/wirecell \
  --with-root=$ROOTSYS \
```

```

--with-libtorch=$LIBTORCH_FQ_DIR \
--boost-includes=$BOOST_DIR/include \
--boost-libs=$BOOST_DIR/lib
# Expected: "Configured for submodules: ..., pytorch, ..., root, ..."
# (must NOT say "Removing package pytorch ...")
./wcb -j$(nproc)
./wcb install
ls $PREFIX/lib/libWireCellPytorch.so    # sanity: the plugin was built

```

Runtime Environment

```

export WIRECELL_FQ_DIR=/home/usher/test/wirecell
export PATH=$WIRECELL_FQ_DIR/bin:$PATH
export LD_LIBRARY_PATH=$WIRECELL_FQ_DIR/lib:$LD_LIBRARY_PATH
export WIRECELL_PATH=$WIRECELL_FQ_DIR/share/wirecell:$WIRECELL_PATH

```

Verification

```

which wire-cell
wire-cell --version
find /home/usher/test/wirecell -name 'libWireCellRoot*'

```

Common Gotchas

- WCT master requires SPDLOG built against EXTERNAL fmtlib. Use `spdlog v1.14.1 + fmt v11.0.2`; `spdlog v1.9.2` (bundled fmt) triggers the `#error`. `pkg-config --cflags spdlog` must show `-DSPDLOG_FMT_EXTERNAL`.
- The reco configs load the WireCellPytorch plugin. If libtorch is not set up at configure time, wcb drops the pytorch package (“Removing package pytorch ... LIB_LIBTORCH”) and jobs die at construction with “failed to load plugin: WireCellPytorch” (art status 9). Set up libtorch and pass `--with-libtorch=$LIBTORCH_FQ_DIR`, then confirm `libWireCellPytorch.so` exists.
- WireCell Jsonnet lookup is not recursive.
- Use explicit relative paths or exact directories in WIRECELL_PATH.
- Do not pass Boost CMake directories to `--boost-libs`.
- Several UPS .pc files may point to stale `/scratch/workspace/...` locations.
- ROOT support is disabled unless `--with-root=$ROOTSYS` is explicitly passed.
- `libWireCellRoot.so` will not build unless ROOT is detected during configure.